Functional & Non-functional Requirements

Supply Chain Performance Analysis: Shipment Delays, Cost Efficiency, Inventory Availability, and the Impact of Shortages on Patient Cases

1. Functional Requirements

ID	Area	Requirement	Reference
FR-01	Data model	Keep two separate data models: shipments (sheet in) and hospitals (inventory_daily, purchase_orders and patient_cases with items, suppliers and hospitals). No relationship exists between the two families.	Project rules; schema_relationships
FR-02	Data model	Relate hospital facts to dimensions with 1:N relationships on validated keys (hospital_id, item_id, supplier_id); link patient_cases to items through key_drug_item_id after standardization.	schema_relationships; Q19
FR-03	Data model	Use one active relationship per fact to the date table when a fact has several date columns; keep the others inactive.	Power BI data model
FR-04	Data layers	Keep an unmodified Raw copy of every source table, a Working copy for changes, and a log of every change with the affected rows.	Raw/Working layers
FR-05	Data preparation	Produce a cleaned dataset with documented derived columns and flags: numeric weight and freight, Delay Days, Schedule Status, Lead Time Days, Customs Clearance Days, Shelf Life at Delivery, PO Delay Days, Stockout Day Flag and standardized categories.	Cleaned Dataset
FR-06	Dashboard	Page 1 Executive Overview (5 visuals): shipment KPI cards, hospital KPI cards, shipment value by year, shipment value by product group, and data completeness by field.	Q01-Q03
FR-07	Dashboard	Page 2 Delivery & Cycle Time (7 visuals): Schedule Status, lead time histogram, median lead time by country, customs clearance against lead time, shelf life histogram, receiving inspection donut, and the table of shipments expired before delivery.	Q04-Q07
FR-08	Dashboard	Page 3 Suppliers & Cost (8 visuals): vendor Pareto, value by product group, unit price by molecule, freight and insurance KPI cards, Freight/Value by country, Freight/Value by product group, freight against weight, and the outlier table.	Q08-Q11
FR-09	Dashboard	Page 4 Purchase Orders & Inventory (9 visuals): actual against historical on-time rate, PO fill ratio, PO delay histogram, PO status donut, lead time against baseline, Stockout heatmap, weekly Stockout line, top 10 items by unmet demand EGP, and wastage by item category and cold chain.	Q12-Q16
FR-10	Dashboard	Page 5 Case Impact & Decision Quality (9 visuals): cases by treatment status, treatment delay box plot, outcomes by treatment status, case table (ICU, complication, length of stay), death share by severity, item stockout against delayed-treatment share, shortage-type share by diagnosis, shortage-type share by hospital, and the decision summary table.	Q17-Q21
FR-11	KPIs	Calculate every analytical KPI defined in the KPI Specification for shipments (13 KPIs) and for the hospital layer (12 KPI groups), each with its documented rules.	KPI Specification
FR-12	Display rules	Show the sample size and the denominator on every visual that shows a rate, a ratio or a median.	KPI rules
FR-13	Display rules	Exclude NULL and invalid values from calculations and show their count; never replace a blank with zero.	Data quality rules
FR-14	Display rules	Group countries with fewer than 5 shipments and vendors with fewer than 3 shipments as Other, and show their count.	Q05, Q08
FR-15	Interaction	Provide slicers on the columns used by each page (for example country, vendor, product group, hospital_id, item category, criticality and treatment_status).	Dashboard pages
FR-16	Interaction	Support cross-filtering, drill-through and tooltips on the five pages.	Functional test
FR-17	Documentation	Provide a data dictionary matched to the source file, a KPI Specification, a README with an execution guide, and a user manual.	Project deliverables

2. Non-functional Requirements

ID	Category	Requirement	Reference
NFR-01	Accuracy	Every final figure matches between SQL, Python, Power BI and the source data; every difference is explained.	Test Plan
NFR-02	Data quality	Data completeness is at least 98% and the duplicate rate is 0% after cleaning (targets marked as assumptions in kpi_definitions); every cleaning step records its affected rows.	kpi_definitions
NFR-03	Data integrity	Blank values remain NULL; outliers are verified and flagged, not deleted because they are large; categories are mapped from existing values only.	Data quality rules
NFR-04	Readability	Neutral background; red and orange for critical indicators and green for good ones; clear labels, sorting, typography and accessibility defined in the dashboard UX specification; every visual answers a stated question and none is misleading.	UX specification
NFR-05	Refreshability	The dashboard refreshes after cleaning without broken relationships or measures.	Final Refresh QA
NFR-06	Traceability	Every result is traceable from source column, to cleaning rule, KPI definition, SQL/Python calculation, Power BI comparison and stated limits.	Execution principle
NFR-07	Validity of interpretation	No causal claims are made; the hospital layer is presented as synthetic; estimates are labeled as estimates.	Project rules
NFR-08	Reproducibility	A team member can run the project from the repository by following the README (installation steps, system requirements - Python 3 with pandas and matplotlib, Power BI Desktop and a SQL engine - configuration and execution guide).	README
NFR-09	Version control	All documents are kept on GitHub in one repository with the folders sql, python, powerbi, data and docs; main branch plus one branch per member; pull request before merge; clear commit messages.	GitHub strategy
NFR-10	Usability	A user understands each dashboard page and each KPI from the user manual without further explanation.	User Manual